INCH-POUND

MIL-W-44480
27 June 1994

MILITARY SPECIFICATION

WAFFLES, SHELF STABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers shelf stable (water activity stabilized) waffles in flexible pouches for use by the Department of Defense as a component of operational rations.

1.2 Classification. The product shall be of the following types, as specified (see 6.1).

Type I - Waffle, plain
Type II - Waffle, berry flavored with added blueberries

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

FEDERAL

A-A-1898 - Cushioning Material, Cellulosic, Packaging

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5018, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8920

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L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin
PPP-C-1752 - Cushioning Material, Packaging, Unicellular, Polyethylene Foam, Flexible
PPP-C-1797 - Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam

STANDARDS

FEDERAL

FED-STD-595 - Colors Used in Government Procurement

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, SW, Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Regulations Governing the Inspection of Eggs and Egg Products (7 CFR 59)

(Copies are available from the Chief, Standardization Branch, Poultry Division, Agricultural Marketing Services, U.S. Department of Agriculture, Room 3944, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS), U.S. FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR 1-199)

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(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AOAC INTERNATIONAL

Official Methods of Analysis of the AOAC

(Application for copies should be addressed to the AOAC International, 2200 Wilson Boulevard, Suite 400, Arlington, VA 22201-3301.)

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

B 479 - Specification for Annealed Aluminum Foil for Flexible Barrier Applications
D 1974 - Standard Practice for Methods of Closing, Sealing, and Reinforcing Fiberboard Shipping Containers
D 3330 - Test Method for Peel Adhesion of Pressure-sensitive Tape at 180° Angle
D 5118 - Standard Practice for Fabrication of Fiberboard Shipping Boxes
F 88 - Test Methods for Seal Strength of Flexible Barrier Material

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, NW, Washington, DC 20418.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-flavors, off-odors, and off-colors.

3.2.1 Baking powder. Baking powder shall be double acting. The baking powder shall be prepared from highly purified food grade materials. The baking powder shall be a white, free-flowing mixture of sodium bicarbonate, cornstarch and one or more of the acid-reacting substances (monocalcium phosphate, sodium acid pyrophosphate or sodium aluminum phosphate). The baking powder shall have an available carbon dioxide level of not less than 13 percent, by weight.

3.2.2 Blueberries, wild, sugar-infused, dried. The dried, sugar-infused, wild blueberries, shall be from properly ripened IQF wild blueberries. The blueberries shall have been infused with high fructose sugar and/or sucrose and then air dried to a moisture of not more than 12 percent. The pH shall be 4.8 (± 0.1), the soluble solids shall be 73 percent (± 2 percent), and the water activity (Aw) shall be 0.580 (± 0.050) (see 6.4.1).

3.2.3 Egg products.

3.2.3.1 Whole eggs, frozen. Whole eggs shall be frozen and shall have been processed and labeled in accordance with the Regulations Governing the Inspection of Eggs and Egg Products (7 CFR 59). The whole eggs shall be egg whites and egg yolks in their natural proportions as broken directly from the shell eggs, as evidenced by a USDA Egg Products Inspection certificate. The frozen, whole eggs shall be USDA certified salmonella free and shall conform to the requirements for Frozen eggs (21 CFR 160.110). The USDA certificate shall also state the date of pasteurization. The USDA certificate shall accompany egg products shipped between plants. Frozen, whole eggs shall be held at 10°F or lower and shall be used within 120 days from the date of pasteurization. The whole eggs shall be free from off-odors or foreign odors such as sour or fruity, and shall be free from foreign colors and materials.

3.2.3.2 Whole eggs, dried. The dried, whole eggs shall be processed and labeled in accordance with the Regulations Governing the Inspection of Eggs and Egg Products (7 CFR 59). The dried, whole eggs shall be USDA certified salmonella free and shall conform to the FDA definition and standard of

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identity for Dried eggs (21 CFR 160.105). The dried, whole eggs shall be used in product preparation within 90 days from date of certification. The USDA certificate shall accompany egg products shipped between plants.

3.2.4 Flavorings.

3.2.4.1 Flavoring, berry. Berry flavoring shall be a liquid or a powder with typical flavor and aroma (see 6.4.2).

3.2.4.2 Flavoring, cream, artificial. Artificial, cream flavoring shall be a liquid or a powder with typical flavor and aroma.

3.2.4.3 Flavoring, vanilla, liquid. Vanilla flavoring shall be a natural or artificial, double strength liquid flavoring.

3.2.5 Flour, cake. Cake flour shall be hard or soft wheat of the type known as high ratio cake flour, bleached, enriched or unenriched. If unenriched flour is used, equivalent enrichment shall be finely incorporated and dispersed by the manufacturer at the time of production of the finished product.

3.2.6 Glycerol. The glycerol shall be food grade, clear, colorless, and odorless and shall comply with the Food Chemicals Codex.

3.2.7 Lecithin. Lecithin shall comply with the Food Chemicals Codex.

3.2.8 Maltodextrin. Maltodextrin shall be a white, free flowing powder with a dextrose equivalent of 10. It shall have a maximum moisture of 6 percent, a pH of 4.0 to 4.7 and have 35 to 40 percent solids.

3.2.9 Oil, vegetable. Vegetable oil shall possess a clear, bland flavor and shall have a minimum stability of 15 hours (A.O.M.), a free fatty acid value not to exceed 0.05 percent, a moisture/volatile matter content not to exceed 0.06 percent and shall pass a cold test of 5.5 hours (minimum).

3.2.10 Potassium sorbate. Potassium sorbate shall be of food grade and shall comply with the Food Chemicals Codex.

3.2.11 Salt. Salt shall be noniodized, white, refined sodium chloride, with or without anticaking agents and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

3.2.12 Starch, instant, granular. The granular, instant starch shall be white to off-white and odorless. It shall have a moisture content of not more than 8.0 percent and the pH shall be in the range of 5.0 to 7.0. A five percent starch solution hydrated in cold or ambient temperature water shall be a smooth, non-grainy paste that develops to a full, spoonable consistency in five minutes. The prepared starch shall display good clarity and a thick, smooth, glossy consistency with good, cold storage stability. It shall be bland with no cereal or starch taste (see 6.4.3).

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3.2.13 Sugar, white, granulated. Sugar shall be white, refined, sucrose, cane or beet sugar, or a combination thereof.

3.2.14 Water. Water used for formulation and washing shall conform to the National Primary Drinking Water Regulations.

3.2.15 Xanthan gum. Xanthan gum shall comply with the Food Chemicals Codex.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Batter formulation. The batter formulation shall be as follows:

Ingredient	Percent by weight	
	Type I	Type II
Flour, cake	33.06	25.56
Water	15.40	17.40
Sugar, white, granulated	15.00	15.00
Eggs, 1/ 2/	13.91	13.91
Oil, vegetable	12.54	12.54
Maltodextrin	4.30	4.30
Glycerol	3.14	3.14
Salt	0.85	0.85
Baking powder	0.68	0.68
Starch, instant, granular	0.50	0.50
Lecithin	0.30	0.30
Flavoring, vanilla, liquid	0.10	0.10
Potassium sorbate	0.10	0.10
Xanthan gum	0.10	0.10
Flavoring, cream, artificial	0.02	0.02
Blueberries, wild, dried, sugar-infused		5.00
Flavoring, berry		0.50

1/ Frozen whole eggs shall be thawed and held at a temperature of 40°F or lower. In no case shall the defrosting period exceed five days. The thawed whole eggs shall be held for not more than 24 hours prior to use in the formula.

2/ Dried, whole eggs may be substituted for whole eggs (liquid basis) by following the manufacturer's recommended rehydration and mixing procedures and shall have no less than the equivalent amount of whole egg solids as the liquid basis.

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3.3.2 Batter preparation. The batter shall be prepared as follows:

a. Combine all dry ingredients together until uniformly distributed.

b. Add liquid ingredients; mix until just combined.

c. When preparing type II waffles, the blueberries shall be added in the final mixing step and mixed until well dispersed.

NOTE: The above preparation procedures were used in the development of the waffles. Alternative procedures may be used provided finished product requirements are met.

3.3.3 Baking. The waffle batter shall be deposited into waffle molds to yield rectangular waffles complying with the finished product requirements. The waffles shall be fully baked to meet all finished product requirements.

3.4 Pouch filling and sealing. Immediately after cooling to a temperature range of 80°F to 120°F, 18 waffles shall be placed into a paperboard tray in accordance with 5.1.1. The paperboard tray with the waffles and FDA approved oxygen scavenger packets shall be immediately inserted into a trilaminate barrier pouch and sealed in accordance with 5.1.

3.5 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insects, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, rancid, scorched, burnt, stale, sour, musty, or moldy.

c. There shall be no color foreign to the product.

d. The waffle have the texture of a fully baked waffle with a light to medium, golden brown crust.

e. The maximum dimensions of each waffle shall be 3-7/16 inches by 2-13/16 inches by 7/8 inch.

f. When bisected vertically with a sharp knife, the waffle shall not crumble, shall not have compression streaks, shall not have gummy centers or soggy areas, shall have no raw portions; shall not be scorched and shall not be uncharacteristically dry and/or crisp.

g. The waffle flavor shall be typical of the labeled identification.

h. Each pouch shall contain 18 intact waffles.

i. No individual pouch shall contain less than 12.0 ounces of product.

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j. The average net weight shall be not less than 12.5 ounces.

k. The water activity for the product in any individual pouch shall be not more than 0.850 (see 4.5.7).

3.5.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.6 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Packing, or Holding of Human Food," and the plant sanitation requirements of the appropriate Government inspection agency.

3.7 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.5.1 Component and material examination. In accordance with 4.1, components and materials shall be examined in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods such as the methods described in the Official Methods of Analysis of the AOAC and in the Approved Methods of the AACC, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.1.2 Laminated pouch material certification. Material listed below may be accepted on the basis of a contractor's certificate of conformance to the specified requirements.

Material requirement	Requirement paragraph	Test procedure
Thickness of films	5.1.1.3	As specified L-P-378 except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378
Aluminum foil thickness	5.1.1.3	As specified in ASTM B 479
Laminated material identification and construction	5.1.1.3	Laboratory evaluation
Color of laminated material	5.1.1.3	Visual evaluation

4.5.1.3 Unfilled preformed pouch certification. A certificate of compliance may be accepted as evidence that unfilled pouches conform to the requirements specified in 5.1.1.3. When deemed necessary by the USDA, testing of the unfilled preformed pouches for seal strength shall be as specified in 4.5.1.3.1.

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4.5.1.3.1 Unfilled preformed pouch seal strength testing. The unfilled pouches shall be tested for seal strength in accordance with ASTM F 88, except that the specimen holding clamps shall be spaced two inches apart prior to testing and the testing speed shall be 10 inches or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. Test specimens shall be cut 1/2 inch or 1 inch in width and to a length suitable for proper mounting. The lot size shall be expressed in pouches. The sample unit shall be one unfilled pouch. The sample size shall be the number of pouches indicated by inspection level S-1. Three adjacent specimens shall be cut from each of the three sealed sides of each pouch in the sample. The results shall be reported to the nearest 0.1 pound per inch of width. The average seal strength of each seal shall be calculated by averaging the strengths of the three test specimens cut from that seal. Any test specimen failing to meet the individual test specimen seal strength requirement or any seal failing to meet the average seal strength requirement specified in 5.1.1.5 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.1.4 Paperboard tray material certification. Components and materials used in the construction of the paperboard tray may be accepted on the basis of a contractor's certificate of conformance to the specified requirements in 5.1.1.1.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation, or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Filled and sealed pouch examination. The filled and sealed pouches shall be examined for the defects listed in table I. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects and 2.5 for minor defects.

TABLE I. Filled and sealed pouch defects 1/

Category		Defect
Major	Minor	
101		Tear, cut, hole, or abrasion through one or more layers in the pouch material or leakage through any heat seal
102		Pouch has foreign odor
103		Not clean 2/
104		Presence of delamination 3/

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TABLE I. Filled and sealed pouch defects 1/ (cont'd)

Category		Defect
Major	Minor	
105		Closure seal width less than 1/16 inch 4/
106		Required labeling missing, incorrect, illegible, or smudged
	201	Closure or top seal extends into or below tear notch
	202	Tear notch(es) missing or not located as specified
	203	Depth of tear notch not as specified
	204	Presence of delamination 3/

1/ Any evidence of insect or rodent infestation shall be cause for rejection of the lot.

2/ Packaging shall be free from foreign matter which is unwholesome, has the potential to cause pouch damage (for example, glass, metal filings,) or generally detracts from the clean appearance of the pouch. The following examples shall not be scored as defects for unclean:

a. Foreign matter which presents no health hazard or potential pouch damage and which can be readily removed by gently shaking the pouch or by gently brushing the pouch with a clean, dry cloth.

b. Dried product which affects less than 1/8 of the total surface area of one pouch face (localized and aggregate).

c. Water spots.

d. Very thin film of grease, oil, or product residue which is discernible to touch but is not readily discernible by visual examination.

3/ Delamination defect classification:

Major: Delamination of the outer ply in the pouch seal area that can be propagated to expose aluminum foil at the food product edge of the pouch after manual flexing of the delaminated area. To flex, the delaminated area shall be held between the thumb and forefinger of each hand with both thumbs and forefingers touching each other. The delaminated area shall then be rapidly flexed 10 times by rotating both hands in

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alternating clockwise, counter-clockwise directions. Care shall be exercised when flexing delaminated areas near the tear notches to avoid tearing the pouch material. After flexing, the separated outer ply shall be grasped between the thumb and forefinger and gently lifted toward the food product edge of the seal, or if the separated area is too small to be held, between the thumb and forefinger, a number two stylus shall be inserted into the delaminated area and a gentle lifting force applied against the outer ply. If separation of the outer ply can be made to extend to the product edge of the seal with no discernible resistance to the gentle lifting, the delamination shall be classified as a major defect. Additionally, spot delamination of the outer ply in the body of the pouch that is able to be propagated beyond its initial borders is also a major defect. To determine if the delaminated area is a defect, use the following procedure. Mark the outside edges of the delaminated area using a bold, permanent marking pen. Open the pouch and remove the contents. Cut the pouch transversely not closer than 1/4 inch (± 1/16 inch) from the delaminated area. The pouch shall be flexed in the area in question using the procedure described above. Any propagation of the delaminated area, as evidenced by the delaminated area exceeding the limits of the outlined borders, shall be classified as a major defect.

Minor: Minor delamination of the outer ply in the pouch seal area is acceptable and shall not be classified as a minor defect unless it extends to within 1/16 inch of the food product edge of the seal. All other minor outer ply delamination in the pouch seal area or isolated spots of delamination in the body of the pouch that do not propagate when flexed as described above shall be classified as a minor defect.

4/ The effective closure seal is defined as any uncontaminated, fusion bonded, continuous path, minimum 1/16 inch wide, from side seal to side seal that produces a hermetically sealed pouch.

4.5.4 Net weight examination. Randomly select 30 filled and sealed pouches from the inspection lot and weigh separately. Determine the average tare weight by randomly selecting and weighing 30 empty paperboard trays, 30 empty pouches, plus the appropriate number of oxygen scavenger packets for one pouch (multiplied by 30) used in packaging the product, and divide the total weight by 30. Subtract the average tare weight from the weight of each filled pouch in the sample. The results shall be reported to the nearest 0.1 ounce. If the net weight of any individual pouch is less than 12.0 ounces, or the average net weight is less than 12.5 ounces, the lot shall be rejected.

4.5.5 Product examination. The filled and sealed sample pouches shall be brought to room temperature (65°F to 75°F) and examined for the defects listed in table II. The lot size shall be expressed in pouches. The sample unit shall be the contents of one pouch. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

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TABLE II. Product defects 1/ 2/

Category		Defect
Major	Minor	
101		Oxygen content in pouch exceeds 0.2 percent 3/
102		Pouch does not contain 18 intact waffles, one paperboard tray, and the appropriate number of oxygen scavenger packets (see 5.1.1.5)
103		When bisected vertically with a sharp knife, the waffle crumbles, has compression streaks, gummy centers or soggy areas; has raw portions; is uncharacteristically dry and/or crisp
104		Tear, hole, or open seal in oxygen scavenger packet
	201	Pouche in which the waffles exceed specified dimensions such that 18 waffles are not packed uniformly and flat in 6 separate stacks of 3 waffles within the paperboard tray
	202	Waffle flavor not typical of the label identification
	203	Crust color not light to medium golden brown

1/ The presence of any foreign material (for example, glass, dirt, insect parts, hair, wood, or metal), foreign odor or flavor (for example, burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability or overall appearance shall be cause for rejection of the lot.

3/ Filled and sealed pouches shall be tested for oxygen content not less than 36 hours after sealing and in accordance with any USDA approved test method. Results shall be reported to the nearest 0.1 percent.

4.5.6 Pouch closure seal testing. The filled and sealed pouches shall be tested in accordance with ASTM F 88, except that the specimen holding clamps shall be spaced 2 inches apart prior to testing and the testing speed shall be 10 inches or 12 inches per minute. Machines that apply tensile load to the specimen by movement of the upper or lower clamp may be used. The test specimens shall be cut to a length suitable for proper mounting. Three adjacent specimens, 1/2 inch or 1 inch wide shall be cut from the closure

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seal of each pouch in the sample. The average seal strength of the closure seal shall be calculated by averaging the test results of the three test specimens cut from that seal. The results shall be reported to the nearest 0.1 pound per inch of width. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The sample size shall be the number of pouches indicated by inspection level S-1. Any individual test specimen or average seal strength failing to meet the requirements of 5.1.1.5 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.7 Water activity testing. Eight filled and sealed pouches shall be randomly selected from each production lot and individually tested for water activity (*Aw*). *Aw* shall be determined in an Agricultural Marketing Service (AMS) laboratory in accordance with the Official Methods of Analysis of the AOAC, method 978.18. Recommendations in sections C, D, E, and F of method 978.18 should be followed for the preparation of reference salt slushes, calibration of instrument and determination of water activity in order to assure accurate water activity measurements. The sample unit shall be one waffle selected from the center of any stack in each pouch. Water activity shall be determined from the center of the sample unit. Water activity testing shall be done at 77°F ± 1°F (25°C ± 1°C). The results of *Aw* determination shall be reported to the third decimal place. Water activity shall be determined not less than 4 days but not more than 14 days after baking to allow moisture equilibration in the product. Any result failing to conforming to the *Aw* requirement in 3.5 shall be classified as major defect and shall be cause for rejection of the lot.

4.5.8 Shipping container examination. Thirteen shipping containers shall be examined for the defects listed below. The finding of any major defect or more than 3 minor defects shall be cause for rejection of the lot.

Major: Cushioning material missing or not as specified.
National stock number, iteme description, contract number, or date of pack markings missing, incorrect, or illegible.
Reinforced with other than nonmetallic strapping or tape.

Minor: More than 30 pounds packed in a box.
Pouches not packed flat in layers.
Other required markings missing, incorrect, or illegible.

5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A.

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5.1.1 Level A. Eighteen waffles on a paperboard tray (see 5.1.1.1) and a sufficient number of oxygen scavenger packets (see 5.1.1.2) to assure compliance with end item oxygen content requirement (see 5.1.1.5) shall be unit packed in a preformed pouch as specified in 5.1.1.3 and 5.1.1.4.

5.1.1.1 Paperboard tray. The paperboard tray shall be food grade and shall be fabricated from solid bleached sulfate paperboard with a thickness of 0.018 inch, and having a basis weight of 198.2 pounds per 3000 square feet. The tray shall have 0.00125 inch thick, food grade polyester extruded on one side. The size of the tray shall be 10-3/8 inches in length, 5-5/8 inches in width, and 1-3/4 inches in depth, and shall be fabricated with the polyester surface on the food contact side.

5.1.1.2 Oxygen scavenger packet. Oxygen scavenger substance shall be FDA approved as suitable for use with foods. The oxygen scavenger shall be packaged in material approved by the FDA for direct contact with food and shall be resistant to the absorption of oil or moisture. The appropriate number of oxygen scavenger packets shall be used to assure that the oxygen content of the gasses in the filled and sealed pouch shall be as specified in 5.1.1.5 (see 6.4.4).

5.1.1.3 Preformed pouch. The preformed pouch shall be fabricated from 0.003 inch to 0.004 inch thick ionomer or polyethylene film laminated or extrusion coated to 0.00035 inch to 0.0007 inch thick aluminum foil which is then laminated to 0.0005 inch thick polyester. The three plies shall be laminated with the polyester on the exterior of the pouch. The complete exterior of the pouch shall be uniformly colored in the range of 34079 through 34087, or 24052 through 24087, or 30045 through 30118 (excluding 30109), or 10045 of FED-STD-595. The material shall be suitably formulated for food packaging (FDA approved) and shall not impart an odor or flavor to the product being packed. The material shall show no evidence of delamination, degradation, or foreign odor when heat sealed or fabricated into pouches.

5.1.1.4 Pouch construction. The preformed pouch shall be a flat style pouch having inside dimensions of 8 inches (± 1/8 inch) by 14 inches (± 1/8 inch). The first dimension specified shall be the opening of the pouch between the heat sealed sides. The pouch shall be made by heat sealing three edges with 3/8 inch (± 1/8 inch) wide seals. The heat seals shall be made in a manner that will assure hermetic seals. The side and bottom seals shall have an average seal strength of not less than 6 pounds per inch and no individual specimen shall have a seal strength of less than 5 pounds per inch when tested as specified in 4.5.1.3.1. A V-shaped or C-shaped (half round) tear notch at least 1/32 inch deep, located 1 to 1-1/4 inches from the top edge of the pouch, shall be made in one or both side seals. The distance between the inside edge of the tear notch and the inside edge of the seal shall be at least 3/16 inch. One side of the open end of the pouch may be provided with an extended or foldover lip, extended not more than 1/8 inch (± 1/16 inch) to facilitate opening and filling. Tear notch location shall be measured from the top of the pouch excluding the extended or foldover lip.

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5.1.1.5 Pouch filling and sealing. Eighteen waffles, 6 stacks of 3 waffles, arranged in a 2 by 3 pattern in a paperboard tray constructed as specified in 5.1.1.1, shall be placed into a pouch constructed as specified in 5.1.1.4. The appropriate number of oxygen scavenger packets to assure that the oxygen content of the gasses in the filled and sealed pouch shall not exceed 0.2 percent, shall be placed into the pouch prior to closure. The filled pouch shall be closed with a continuous heat seal not less than 1/4 inch wide. If thermal impulse or combination (heated curved bar with thermal impulse) sealing is used, any seal width from 1/8 to 7/16 inch will be acceptable. The closure seal shall not extend below the tear notch on either side of the pouch. The closure seal shall be free of foldover wrinkles and delaminations. The average seal strength shall be not less than 6 pounds per inch, and no individual test specimen shall be less than 5 pounds when tested as specified 4.5.6.

5.2 Packing for shipment to ration assembler. Not more than 30 pounds of packaged product shall be packed flat in a fiberboard box conforming to style RSC-L, class domestic, variety SW, grade 200 of ASTM D 5118. Cushioning material not less than 1/8 inch thick conforming to grade II, class B or C of A-A-1898, or cushioning material not less than 1/16 inch thick, conforming to type I of PPP-C-1797, or cushioning material conforming to type VII, class 1 of PPP-C-1752 shall be placed between each of the layers and in the bottom and top of the box to minimize movement of the individual packages. Each shipping container shall be closed in accordance with ASTM D 1974.

5.3 Labeling and marking.

5.3.1 Pouches. Each pouch shall be clearly printed with permanent ink in large letters of black, purple, or similar dark contrasting color with the following information:

WAFFLES, SHELF STABLE

INGREDIENTS:

TO SERVE

Waffles may be heated prior to serving or served at ambient temperature without heating. When heated, the following instructions shall apply:

To heat in water: Place unopened pouch in simmering water for five minutes.

To heat in oven: Remove pouch and place waffles in an oven at 250°F for five minutes. DO NOT OVERHEAT.

YIELD: Serves 6 portions, 3 waffles per serving.

NET WEIGHT:

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NAME AND ADDRESS OF PRODUCER:

DATE OF PACK (expressed as a four digit Julian code):

5.4 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type of product required (see 1.2)
c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.4, and 6.2).
e. Provisions for approved preproduction samples (see 3.5.1 and 6.2).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.4 Ingredient information.

6.4.1 Blueberries, wild, sugar-infused, dried. It has been found that Wild Blueberries, Dried, produced by Wyman's Food Products, P.O. Box 100, Milbridge, ME 04658. meet the requirements of 3.2.2 and perform satisfactorily in this product.

6.4.2 Flavoring, berry. It has been found that Natural and Artificial Wild Berry Flavor, #096933, produced by F & C International Inc., Cincinnati, OH 45215, meets the requirements of 3.2.4.1 and performs satisfactorily in this product.

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6.4.3 Starch, instant, granular. It has been found that MIRA-SPERSE, produced by A.E. Staley Co. meets the requirements of 3.2.12 and performs satisfactorily in this product.

6.4.4 Oxygen scavenger packets. It has been found that oxygen scavenger packets manufactured by Multiform Desiccants, Inc., Buffalo, NY 14213 meets the requirements of 5.1.1.2 and performed satisfactorily in the development of this product. Other approved oxygen scavenger packets may be used.

6.5 Subject term (key word) listing.

Combat field feeding
Operational rations
Food processing

Custodians:

Army - GL
Navy - SA
Air Force - 35

Preparing activity:

Army - GL

(Project 8920-0563)

Review activities:

Army - MD, QM
Navy - MC
DLA - SS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-W-44480	1994 JUNE 27

3. DOCUMENT TITLE
WAFFLES, SHELF STABLE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

[redacted]

7. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4501	256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WRE Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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